Sequential Games

Key idea

- Lectures 1-2 : each player makes only one choice, and all choose simultaneously.
- This week : players make (one or multiple) choices **sequentially**, as in chess.
- Main message : the Nash equilibrium solution concept is **not fully satisfactory** for sequential games. We refine it with the **Subgame-perfect equilibrium**.

Definition

SEQUENTIAL GAME
Needs to specify :

- the players;
- the order of moves (which player moves at time 0, 1, 2…?);
- the possible moves of each player at each of her decision points;
- the information available to players throughout the game.

Definition

PERFECT INFORMATION
At each time 0, 1, 2, … each player knows the actions taken by the other players up to that time (assumed in this lecture).

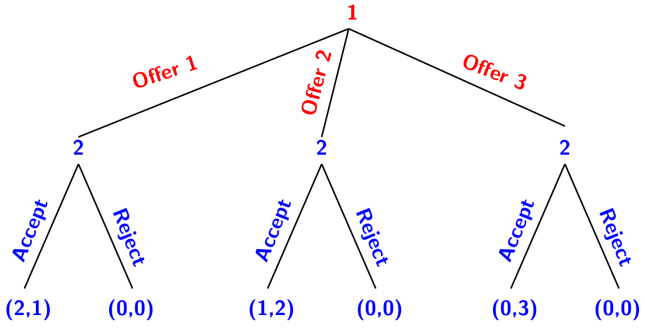
Example : Ultimatum Game

Example

- Players : 1 and 2.
- Player 1 has 3 coins, but needs to share them with Player 2.
- First, Player 1 makes an offer to Player 2 : he can offer 1, 2 or 3 coins.
- Next, Player 2 can accept or reject the offer. If he accepts, he gets the n (1, 2 or 3) coins offered and Player 1 keeps the rest. If Player 2 rejects, nobody gets anything.

Definition

EXTENSIVE FORM
Sequential games are best represented in extensive form (= as **decision trees**).



Pure Strategies and Normal Form

Definition

PURE STRATEGY (SEQUENTIAL GAME)
A pure strategy for Player i in a sequential game of perfect information is a **contingent plan** that specifies an action at **each decision node** of Player i.

Example

Ultimatum game : a pure strategy for Player 1 could be simply "Offer 1". A pure strategy for Player 2 could be "Reject if the offer is 1 or 2, and accept if the offer is 3".

Normal-form representation of the Ultimatum game

$1\backslash 2$	RRR	RRA	RAA	AAA	ARR	RAR	AAR	ARA
Offer 1	(0,0)	(0,0)	(0,0)	(2,1)	(2,1)	(0,0)	(2,1)	(2,1)
Offer 2	(0,0)	(0,0)	(1,2)	(1,2)	(0,0)	(1,2)	(1,2)	(0,0)
Offer 3	(0,0)	(0,3)	(0,3)	(0,3)	(0,0)	(0,0)	(0,0)	(0,3)

The eight choices for Player 2 are his eight possible pure strategies : e.g. 'RRR' means Reject if offered 1, 2 or 3; 'RAR' means Reject if offered 1, Accept if offered 2, Reject if offered 3.

Nash Equilibrium of Sequential Games

Definition

NASH EQUILIBRIUM
Defined the same way as for simultaneous-move games : a strategy profile $(s^{(1)}, s^{(2)}, \dots, s^{(N)})$ is a Nash equilibrium iff

$$\mathcal{P}^{(i)}(s^{(i)}, s^{(-i)}) \geq \mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) \quad \text{for every possible } \tilde{s}^{(i)}$$

i.e. the best that one player can do *given the strategies of all the other players*.

Example

The Ultimatum game has many pure-strategy NE (underlined) :

$1\backslash 2$	RRR	RRA	RAA	AAA	ARR	RAR	AAR	ARA
Offer 1	(0,0)	(0,0)	(0,0)	(2,1)	(2,1)	(0,0)	(2,1)	(2,1)
Offer 2	(0,0)	(0,0)	(1,2)	(1,2)	(0,0)	(1,2)	(1,2)	(0,0)
Offer 3	(0,0)	(0,3)	(0,3)	(0,3)	(0,0)	(0,0)	(0,0)	(0,3)

Take for example the NE (Offer 2, RAA) : Player 2 will not accept unless he is offered at least 2. Player 1 knows it, and therefore offers 2.

Non-credible Threats

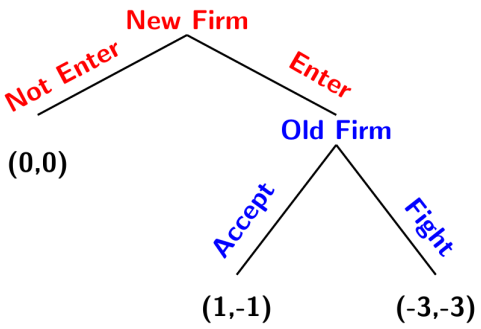
Warning

The NE (Offer 2, RAA) involves a **non-credible threat** by Player 2 : that he will Reject (and therefore get nothing) if he is offered 1.
But once Player 1 offers 1, it would be irrational for Player 2 to reject the offer and walk away with nothing!

The Entry-Deterrence Game

Definition

STRATEGIC ENTRY DETERRENCE
In business, any action taken by an existing business in a particular market that discourages potential entrants from entering into competition in that market (barriers to entry : hostile takeovers, heavy spending on new product development, predatory pricing…).



New Firm \ Old Firm	Accept	Fight
Not Enter	(0,0)	(0,0)
Enter	(1,-1)	(-3,-3)

NE underlined.

Key idea

Two pure-strategy NE : **(Not Enter, Fight)** and **(Enter, Accept)**.
But (Not Enter, Fight) is again a **non-credible threat**.

Sequential Rationality

Definition

SEQUENTIAL RATIONALITY
A player's equilibrium strategy should specify **optimal actions at every decision node** in a tree.

Example

— **Entry deterrence** : Fight is a suboptimal decision in the decision node that starts after New Firm chooses 'Enter'. Therefore, the NE (Not Enter, Fight) is not realistic.
— **Ultimatum game** : the NE (Offer 2, RAA) – and also (Offer 2, RAR) and (Offer 3, RRA) – do not satisfy the Sequential Rationality principle either.

Key idea

We need a refinement of the concept of Nash equilibrium for a sequential game, that satisfies sequential rationality.

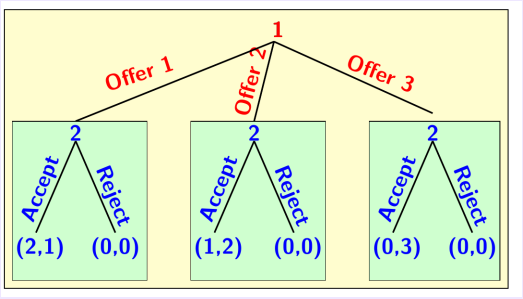
Subgames

Definition

SUBGAME
Given a game G, a subgame G_1 is the portion of the game G starting at one of the decision nodes and containing all its successor nodes.

Example

Subgames of the Ultimatum game : the whole game (yellow) and the three subgames starting at the decision nodes of Player 2 (green).



Subgame-Perfect Equilibrium (SPE)

Definition

SUBGAME-PERFECT EQUILIBRIUM
A subgame-perfect equilibrium of an extensive-form game G is a strategy profile s such that for any subgame G', the restriction of s to G' is a Nash equilibrium for G'.

Key idea

A subgame-perfect equilibrium is a special kind of Nash equilibrium (obvious since G is a subgame of itself).

Backward Induction

Summary

Backward induction – the technique to obtain a Subgame-Perfect equilibrium :

1. Find the Nash equilibria of the smallest subgames.
2. Replace each subgame with the payoff of (one of) its NE.
3. Repeat the procedure with the reduced game.

Example

Entry-Deterrence game

— The smallest subgame (after 'Enter') has only one player, Old Firm, who will choose 'Accept'. We replace this subgame with the payout of (Enter, Accept) : $(1, -1)$.
— The reduced game (Not Enter : $(0, 0)$, (Enter, A) : $(1, -1)$) has only itself as a subgame. It is a one-player game (New Firm), who will choose Enter.

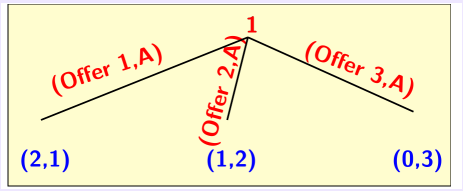
SPE : **(Enter, Accept)**.

Example

Ultimatum game

— In each of the smallest subgames there is only Player 2, and in each of these the only NE is 'Accept'.

— After replacing them with the payout of their NE, we obtain the reduced game :



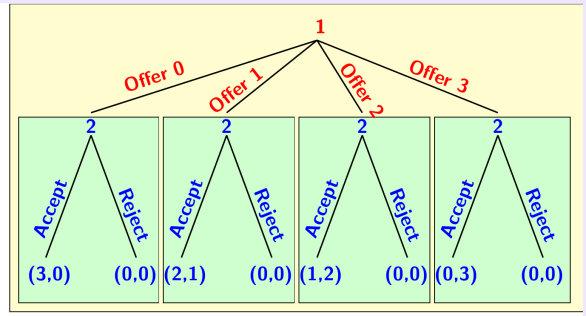
— Only Player 1 needs to choose, and will choose 'Offer 1'.

SPE : **(Offer 1, AAA)**.

An Example with two SPE

Example

New version of the Ultimatum Game : Player 1 still has 3 coins, but can now also offer 0 coins to Player 2 (so can offer 0, 1, 2, or 3 coins).



— In the subgame starting after 'Offer 0', Player 2 gets 0 whether he accepts or rejects : 'Accept' and 'Reject' are **two pure-strategy NE** of this subgame.

— In the other three subgames, the only NE is 'Accept'.

Key idea

Solution 1 : choose the NE 'Accept' for all 4 subgames. Reduced game : (Offer 0,A) $(3,0)$, (Offer 1,A) $(2,1)$, (Offer 2,A) $(1,2)$, (Offer 3,A) $(0,3)$. Player 1 chooses 'Offer 0', which gives her the highest payout (3).
SPE : **(Offer 0, AAAA)**.

Solution 2 : choose the NE 'Reject' for the subgame starting after 'Offer 0', and the (only) NE 'Accept' for the other 3 subgames. Reduced game : (Offer 0,R) $(0,0)$, (Offer 1,A) $(2,1)$, (Offer 2,A) $(1,2)$, (Offer 3,A) $(0,3)$. Player 1 chooses 'Offer 1', which gives her the highest payout (2).
SPE : **(Offer 1, RAAA)**.

Threat Credibility : Cuban Missile Crisis (1962)

Threat credibility is very important in politics and international relations. Game theory tools have been used to analyze the **Cuban missile crisis in 1962** : one of the most dramatic points in the Cold War, unfolding over 13 days, in which many feared that a nuclear war might be approaching.

Example

— The USA, led by President JFK, discovered that **the USSR was in the process of installing missiles in Cuba**, to deter US attacks to the island and more generally to provoke and threaten the USA.

— The main "punishments" the USA discussed were a **naval blockade to Cuba** and an **air-strike to destroy the missiles**.

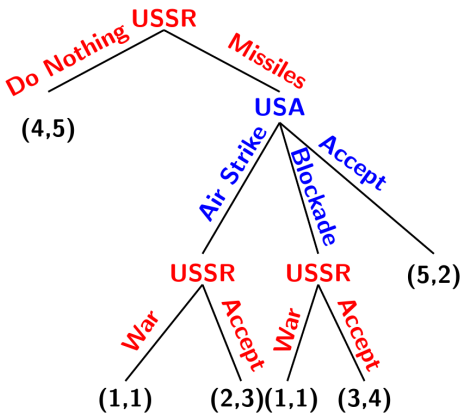
— The USSR, hoping that the USA would just accept the missiles without retaliation, made it known that both these options would be acts of war, that would **inevitably lead to a nuclear war between USA and USSR**.

Order of preference over final outcomes
USSR :

1. Missiles installed, USA accepts without retaliation
2. Status Quo (No Missiles installed)
3. Missiles installed, USA retaliation through naval blockade, no war
4. Missiles installed, USA retaliation through air strike, no war
5. Nuclear War

USA :

1. Status Quo (No Missiles installed)
2. Missiles installed, USA retaliation through naval blockade, no war
3. Missiles installed, USA retaliation through air strike, no war
4. Missiles installed, USA accepts without retaliation
5. Nuclear War



Warning

Different payouts preserving the same preference ranking for USA and USSR would lead to the same NE and SPE.

Normal form and NE (underlined). USSR strategies : first letter = first move (N : Do Nothing, M : Missiles), then the answer (W : War, A : Accept) after an Air Strike, then after a Blockade.

USSR \ USA	Air Strike	Blockade	Accept
NWW	(4,5)	(4,5)	(4,5)
NWA	(4,5)	(4,5)	(4,5)
NAW	(4,5)	(4,5)	(4,5)
NAA	(4,5)	(4,5)	(4,5)
MWW	(1,1)	(1,1)	(5,2)
MWA	(1,1)	(3,4)	(5,2)
MAW	(2,3)	(1,1)	(5,2)
MAA	(2,3)	(3,4)	(5,2)

Key idea

Of all the NE, only **(NAA, Blockade)** is a SPE.
The USSR was counting on the USA to believe that the MWW strategy would be credible, so that the realized NE would be (MWW, Accept). Eventually, the USSR agreed to remove the missiles, consistently with the (NAA, Blockade) SPE.

Weekly Summary

Summary

- Sequential games are represented in extensive form (decision trees).
- A pure strategy is a contingent plan : an action at each decision node of the player.
- The NE of sequential games can rely on **non-credible threats**.
- **Sequential rationality** : optimal actions at every decision node.
- **SPE** : the restriction to any subgame is a NE of that subgame.
- SPE are found by **backward induction** ; a game can have several SPE.
- If we have time : the “pirate game” video, a 5-person game solved by backward induction (see week 04).